

CRYPTOGRAPHY

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Exam: Written. Consists 3 parts; each part
1 exercise + 1 theory question.

Main Textbook: "Introduction To Modern Cryptography".
J. Katz, Y. Lindell.

What The course is about.

We'll cover the basics of cryptography,
with a focus on PROvable SECURITY.

The modern approach to crypto; The field
has changed dramatically since 1950.

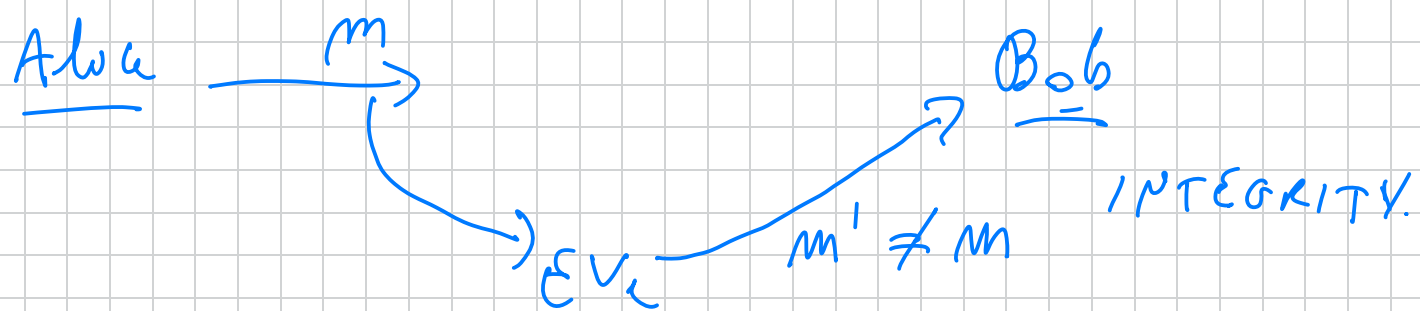
< 1950: Trial & error

≥ 1950: Science.

Provable security: We should have precise definitions of what secure
means. Then also a security proof (under maybe some assumptions).

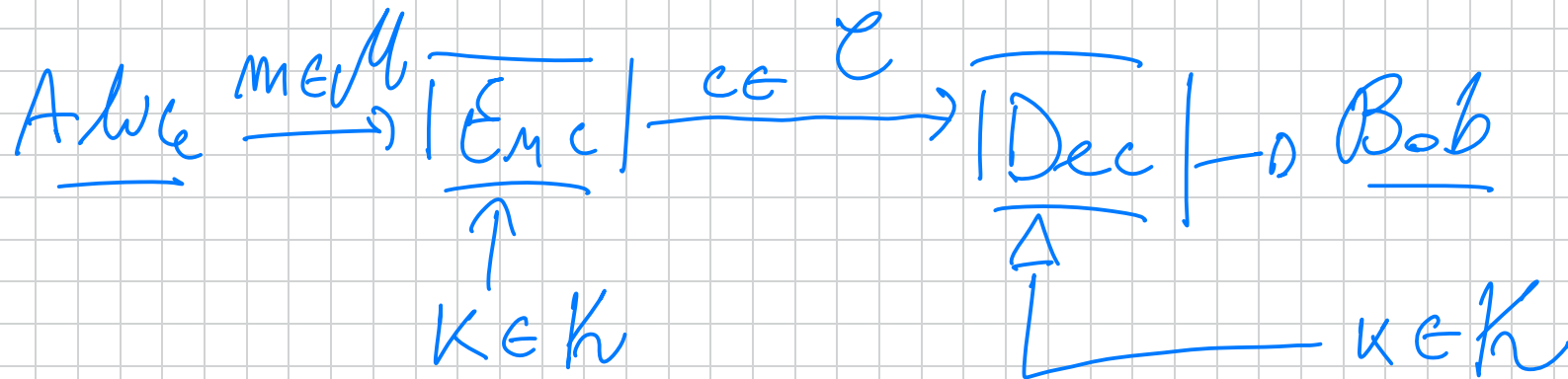
More in details, we will aim for SECURE COMMUNICATION.





UNCONDITIONAL SECURITY

Treatment due to CLAUDE SHANNON.



$$\Pi = (\text{Enc}, \text{Dec})$$

Assumption: $K \in \mathcal{K}$ is uniformly random.
(unknown to Eve)

$$\text{Enc}: \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{C}$$

$$\text{Dec}: \mathcal{K} \times \mathcal{C} \rightarrow \mathcal{M}$$

CORRECTNESS: $\forall k \in \mathcal{K}, \forall m \in \mathcal{M}$

$$\text{Dec}(k, \text{Enc}(k, m)) = m$$

DEF (PERFECT SECRECY). Let M be any distribution on $\mathcal{M}$, and K the uniform dist. on $\mathcal{K}$.

Then, $C = \text{Enc}(K, M)$ is a distribution

We say that $\Pi = (\text{Enc}, \text{Dec})$ is perfectly secret if $\forall M, \forall m \in \mathcal{M}, \forall c \in \mathcal{C}$

$$\Pr[M = m] = \Pr[M = m \mid C = c].$$

Next: We'll show That PERFECT SECRECY is possible, but at a high price.

THM. The following are equivalent:

(i) Perfect secrecy.

(ii) M, C are INDEPENDENT ($I(M, C) = 0$)

(iii) $\forall m, m' \in \mathcal{M}, \forall c \in \mathcal{C}$

$$\Pr[E_{mc}(K, m) = c] = \Pr[E_{mc}(K, m') = c]$$

Let's first apply the theorem.

ONE-TIME PAD

$$K, \mathcal{M}, \mathcal{C} = \{0, 1\}^l$$

$$\text{Enc}(K, m) = K \oplus m$$

$$\text{Dec}(K, c) = K \oplus c$$

$$= K \oplus (K \oplus m) = m \quad \checkmark$$

Cor One-Time pad is PERFECTLY SECRET.

Proof. For any $m \in \mathcal{M}$, $c \in \mathcal{C}$

$$\Pr[\text{Enc}(K, m) = c] =$$

$$\Pr[K \oplus m = c] = \Pr[K = c \oplus m]$$

$$\text{For } m' \in \mathcal{M}: \quad = 2^{-l}$$

$$\Pr[\text{Enc}(K, m') = c] = 2^{-l} \quad \square$$

The Properties:

1) length of K same as length of m .

2) The key can only be used once.

For OTP: $C_1 = K \oplus M_1$

$$C_2 = K \oplus M_2$$

$$C_1 \oplus C_2 = M_1 \oplus M_2$$

$\oplus$	0	1
0	0	1
1	1	0

$$K \oplus K = 0^l$$

Proof (of THP).

$(N) \Rightarrow (NN)$

$$\begin{aligned} \Pr[M=m] &= \Pr[M=m \mid C=c] \\ &= \frac{\Pr[M=m \wedge C=c]}{\Pr[C=c]} \end{aligned}$$

$$\Rightarrow \Pr[M=m \wedge C=c] = \Pr[M=m] \cdot \Pr[C=c]$$

$(NN) \Rightarrow (NNN)$

$$\Pr[E_{mc}(K, m) = c] =$$

$$\Pr[E_{mc}(K, M) = c \mid M = m] =$$

$$\Pr[C=c \mid M=m] =$$

$$\Pr[C=c].$$

By the same steps:

$$\Pr[E_{mc}(K, m')=c] = \Pr[C=c]$$

(iiv) $\Rightarrow$ (v).

We will first show $\Pr[C=c \mid M=m] = \Pr[C=c]$

$$\Pr[C=c] = \sum_{m'} \Pr[C=c \wedge M=m']$$

$$= \sum_{m'} \Pr[C=c \mid M=m'] \cdot \Pr[M=m']$$

$$= \sum_{m'} \Pr[E_{mc}(K, M)=c \mid M=m'] \cdot \Pr[M=m']$$

$$\begin{aligned}
&= \sum_{m'} \Pr[E_{mc}(K, m') = c] \cdot \Pr[M = m'] \\
&= \sum_{m'} \Pr[E_{mc}(K, m) = c] \cdot \Pr[M = m'] \\
&= \Pr[E_{mc}(K, m) = c] \cdot \underbrace{\sum_{m'} \Pr[M = m']}_{=1}
\end{aligned}$$

$$\begin{aligned}
&= \Pr[E_{mc}(K, M) = c \mid M = m] \\
&= \Pr[C = c \mid M = m]
\end{aligned}$$

$$\text{So, } \Pr[C = c] = \Pr[C = c \mid M = m]$$

By Bayes:

$$Pr[M=m | C=c] \cdot Pr[C=c] =$$

$$Pr[M=m \wedge C=c] =$$

$$Pr[C=c | M=m] \cdot Pr[M=m]$$

$$\Rightarrow Pr[M=m] = \frac{Pr[M=m | C=c] \cdot \cancel{Pr[C=c]}}{Pr[C=c | M=m]} \\ = \cancel{Pr[C=c]} \quad \square$$